Fiscal Year:	FY 2024	**Task Last Updated:**	FY 04/17/2025
PI Name:	Mohraz, Ali Ph.D.		
Project Title:	Bijel Processing: Decoupling the Roles of Marangoni and Gravitational Forces		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mohraz@uci.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	949-824-2028
Organization Name:	University of California, Irvine		
PI Address 1:	Chemical Engineering and Materials Science		
PI Address 2:	916 Engineering Tower		
PI Web Page:			
City:	Irvine	**State:**	CA
Zip Code:	92697-2575	**Congressional District:**	45
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	04/13/2021	**End Date:**	12/08/2023
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
Contact Email:	david.f.chao@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS Space X NOTE: End date changed to 12/08/2023 per NSSC information (Ed., 11/2/22).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC21K0897		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This is ongoing university science support for Advanced Colloids Experiments (ACE) colloids microgravity research experiment on the International Space Station (ISS). Bicontinuous interfacially jammed emulsion gels (bijels) are a new class of soft materials with interpenetrating co-continuous arrangement of two immiscible liquid phases separated by a monolayer of colloidal particles. Because of this unique morphology, bijels offer significant technological potential for advanced materials synthesis with applications in electrochemical energy storage and human health applications, with initial efforts along this realm already underway. To realize their full technological utility, a more thorough understanding of the physicochemical and morphological factors that influence the mechanical stability and processability of bijels is required. In particular, Marangoni forces that result during bijel processing, as well as gravitational forces acting to separate the different ingredients in these inherently multiphase materials by density, both work against bijel stability and processability. Because of the proximity of bijel physicochemical conditions to phase boundaries, temperature provides a robust handle for investigating the role of Marangoni forces in bijel physics. However, these effects are then intimately coupled to gravitational separation, and cannot be separately studied on the Earth. Using synergistic ground-based and microgravity experiments and theoretical calculations, the proposed program aims to develop quantitative metrics for the assessment of bijels stability in the presence of external forces such as gravity and surface tension gradients during bijel processing. Success of the proposed program would further enable the design and synthesis of novel co-continuous microstructured composites for applications in advanced manufacturing, energy, nanotechnology, and health care.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Bicontinuous interfacially jammed emulsion gels (bijels) have a unique microstructure that is suitable for the development of advanced materials and composites that require many sample-spanning phases to co-exist. For example, next generation battery electrodes are envisioned as 3D arrangements of ion- and electron-conducting phases in coexistence with electrolyte channels that percolate throughout the entire electrode volume. This morphology can be uniquely realized in bijels, as they inherently enjoy a co-continuous microstructure that is self-assembled through an external stimulus (temperature change). Similarly, the co-continuous morphology of bijel-derived materials can prove beneficial for tissue engineering and cell delivery scaffolds or any other structure that requires microbial growth, such as electrodes in microbial fuel cells. Finally, the open pore morphology and uniform domain size uniquely realized in bijels can lead to better hydraulic permeability for a given porosity, leading to enhanced transport and lower energy consumption for fluid flow. Therefore, bijel processing will provide a unique route to advanced energy and health care materials, and understanding the rheology and processability of bijels, which is the focus of this project, will pave the way for the realization of this end goal. Applications of bijels in each of these areas are currently being investigated in our laboratory.
Task Progress:	The overarching objective of this research program was to assess how Marangoni forces impact the processability of bijels (bicontinuous interfacially jammed emulsion gels). We aimed to investigate the relationship between the microstructure and rheology of this class of soft multiphase materials, while also investigating and expanding the Earth benefits of bijel-derived composites. In addition, we sought new bijel chemistries to enable safe handling and experiments with bijels at the International Space Station (ISS), and long-duration rheology experiments on Earth. A series of experiments were planned to 1) expand the chemical library of bijels and bijel-derived materials; and 2) help uncover how various forces at the colloidal scale impact the stability of bijels through a combination of ground and microgravity experiments, as well as to develop basic theoretical models of bijel rheology. The major activities during the award period involved the development of a new method to synthesize bijel-derived materials more rapidly; development of new bijel chemistries that enabled long-duration rheological characterization of bijels; a full experimental investigation of the rheology of bijels in the linear and nonlinear regimes; development of a basic model to describe bijel rheology beyond the limit of linear viscoelasticity; and experiments aboard the ISS to understand the effect of Marangoni forces on bijels. Our efforts and findings on each front formed the bases of two peer-reviewed manuscripts that were published in Materials Advances and Soft Matter. [Ed. Note: See Bibliography.] One additional manuscript is to be submitted shortly.
Bibliography Type:	Description: (Last Updated: 04/22/2025)
Articles in Peer-reviewed Journals	Ching H, Mohraz A. "A basic model for the nonlinear rheology of bijels." J. Rheol. 2025 Jan 1;69(1):45-54. https://doi.org/10.1122/8.0000914 , Jan-2025
Articles in Peer-reviewed Journals	Gonthier AR, Botvinick EL, Grosberg A, Mohraz A. "Effect of porous substrate topographies on cell dynamics: A computational study." ACS Biomater. Sci. Eng. 2023 Sep 15;9(10):5666-78. https://doi.org/10.1021/acsbiomaterials.3c01008 , Sep-2023
Articles in Peer-reviewed Journals	Ching H, Mohraz A. "Bijel rheology reveals a 2D colloidal glass wrapped in 3D." Soft Matter. 2022 May 17;18(22):4227-38. http://dx.doi.org/10.1039/D2SM00407K , May-2022
Articles in Peer-reviewed Journals	Gross SJ, Hsieh MT, Mumm DR, Valdevit L, Mohraz A. "Alleviating expansion-induced mechanical degradation in lithium-ion battery silicon anodes via morphological design." Extreme Mechanics Letters. 2022 Jul 1;54:101746. https://doi.org/10.1016/j.eml.2022.101746 , Jul-2022